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CSI-Feedback-Driven Link Adaptation in 5G NR Downlink MIMO: A Link- Level Simulation Study

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I hereby declare that, except where specifically indicated, the work submitted herein is my own original work.

Signed Kaiyuan Bao 包铠源 date 30 May 2026

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The completion of this project marks the end of my four-year study at Cambridge. I would like to take this opportunity to thank my parents for their tremendous support during my years in the UK, which made it possible for me to pursue my studies at Cambridge.

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Technical Abstract

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Background and Motivation

Methodology

Key Results and Interpretation

Conclusion

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List of Abbreviations

Abbreviation	Definition
5G	Fifth Generation
BLER	Block Error Rate
CDL	Clustered Delay Line
CP	Cyclic Prefix
CQI	Channel Quality Indicator
CSI	Channel State Information
CSI-RS	CSI Reference Signal
DL	Downlink
DM-RS	Demodulation Reference Signal
FR	Frequency Range
gNB	Next Generation Node B
HARQ	Hybrid Automatic Repeat Request
LDPC	Low-Density Parity-Check
LOS	Line-of-Sight
MCS	Modulation and Coding Scheme
MIMO	Multiple-Input Multiple-Output
NLOS	Non-Line-of-Sight
NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing
PDCCH	Physical Downlink Control Channel
PDSCH	Physical Downlink Shared Channel
PMI	Precoding Matrix Indicator
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RB	Resource Block
RE	Resource Element
RI	Rank Indicator
SCS	Subcarrier Spacing
SINR	Signal-to-Interference-and-Noise Ratio
SISO	Single-Input Single-Output
SNR	Signal-to-Noise Ratio
TDL	Tapped Delay Line
UE	User Equipment
UL	Uplink

1 Introduction

1. 5G NR, Physical Layer, MIMO background. Why 5G needs mimo and adaptive transmission
2. Importance of CSI
3. Why fixed-rank, fixed-modulation BER curve is not enough 3. Matlab 5G toolbox simulation
4. Baseline scenario
5. RQs
6. Structure of my research

Mobile telecommunications have experienced rapid development over the past two decades. Starting from the deployment of third-generation (3G) systems in the early 2000s, followed by the introduction of fourth-generation long-term evolution (4G LTE) in the early 2010s, each generation has delivered substantial improvements in data rate and user experience. For end users, the most noticeable change has been the transition from limited and slow web browsing to smooth video streaming and real-time online services. These improvements were enabled by key physical-layer innovations, including the use of orthogonal frequency division multiple access (OFDMA) in 4G LTE, replacing the code division multiple access (CDMA) used in earlier systems, as well as the introduction of more advanced multi-antenna techniques [1].

Building upon the capabilities of 4G, the fifth-generation mobile communication system (5G) has been designed to further increase data rates, reduce latency, and support a wide range of services. Three main usage scenarios are defined for 5G: enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultra-reliable and low-latency communications (URLLC) [2]. Together, these scenarios impose diverse requirements on 5G, ranging from extremely high data rates to massive device connectivity, as well as stringent latency and reliability constraints. To meet these requirements, multiple-input multiple-output (MIMO) techniques play a central role. In particular, massive MIMO systems enable significant throughput gains through spatial multiplexing, null forming and beamforming [3].

The realization of these requirements relies on globally agreed technical specifications, which are defined by the Third Generation Partnership Project (3GPP), a global standardisation body established in 1998 for the development of specifications for 3G. Its scope has expanded from third-generation of cellular networks to the standardisation and continuous evolution of 4G, 5G, and technologies beyond. The first release of 5G NR, Release 15, was published by 3GPP in 2019, marking the point at which 5G NR became ready for practical implementation. In the release, core technical specifications for 5G NR were defined. 3GPP is currently continuing its work on 5G-Advanced and early studies towards 6G [4].

This research project aims to provide insights that can support future 3GPP standardisation discussions related to MIMO techniques in 5G and beyond. Simulations are conducted in MATLAB using a framework aligned with 3GPP specifications. This alignment includes the use of standardized channel models as well as baseline system parameters commonly adopted in 3GPP testings, such as carrier frequency and bandwidth. This report introduces the relevant background concepts, presents the adopted simulation methodology, and summarises the current progress and preliminary findings of the project.

2 Background and Methodology

2.1 5G NR Downlink Physical Layer Overview

1. OFDM
2. PDSCH (DL)
3. DM-RS
4. CSI-RS
5. Slot, RB
6. Definition of BER/BLER/Throughput
7. Why throughput not BER

2.2 MIMO Transmission and CSI Feedback

Specifics of CSI: PMI, RI, CQI

Feedback loops

How the modulation scheme/TCR/rank changes based on the feedback

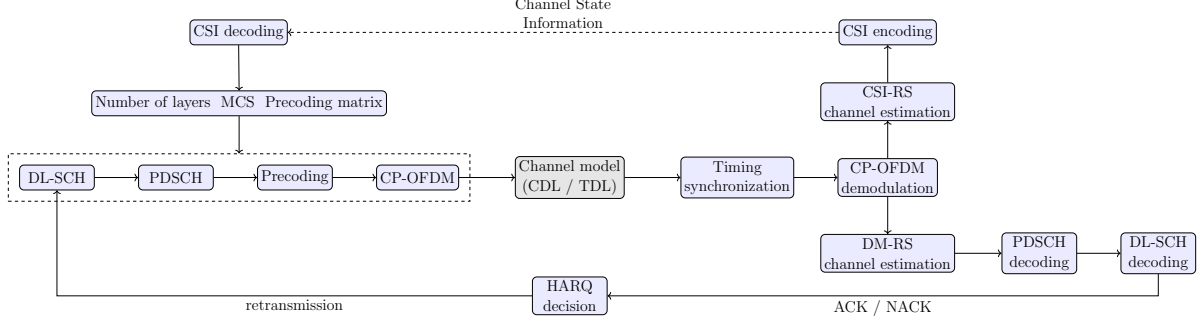


Figure 1: Overview of the CSI-based downlink transmission and feedback framework, adapted from [5]

As illustrated in Figure 1, the downlink transmission relies on channel state information (CSI) obtained through reference signal measurements at the receiver. The user equipment (UE) estimates the downlink channel based on CSI reference signals (CSI-RS). Based on the estimated channel, the CSI is encoded into three main feedback components: the Channel Quality Indicator (CQI), Precoding Matrix Indicator (PMI), and Rank Indicator (RI).

CQI is a 4-bit indicator with values ranging from 0 to 15, representing the reported downlink channel quality. It provides an indication of the highest modulation scheme and coding rate that can be supported under a predefined target block error rate (BLER). The precoding matrix indicator (PMI) specifies the precoding matrix applied at the transmitter, while the rank indicator (RI) indicates the recommended number of transmission layers [5].

There are various parameters that affect the quality of the CSI. These include the CSI reporting periodicity, which controls how frequently the feedback is updated, as well as the reporting mode, which can be configured as wideband or subband. In addition, parameters such as the codebook type and mode, and the subband size further impact the granularity of the reported CSI. In time-varying channels, infrequent or coarse CSI reporting may lead to outdated channel knowledge at the transmitter, therefore degrading the performance of the transmission.

The reported CSI is fed back to the transmitter and used to configure key downlink transmission parameters, including the modulation scheme, target code rate, number of transmission layers, and the precoding matrix. Using these updated parameters, the transmitter processes the data through the DL-SCH and maps it to the PDSCH, followed by precoding and CP-OFDM modulation before transmission over the wireless channel. At the receiver, the received signal is synchronised, demodulated, and decoded using demodulation reference signals (DM-RS) for coherent detection.

There are two widely used channel models in 5G NR link-level evaluations, namely the tapped delay line (TDL) model and the clustered delay line (CDL) model. Each model has several variants designed to represent different propagation environments such as urban, suburban, and rural scenarios. The TDL model provides a simplified representation of the multipath channel and is commonly used for baseline evaluations or scenarios where spatial characteristics and multi-antenna effects are not modelled. In the TDL model, the channel is represented as a superposition of multiple discrete propagation paths (taps), each characterised by a specific delay τ_l and a path gain $h_l(t)$:

$$h(t, \tau) = \sum_{l=0}^{L-1} h_l(t) \delta(\tau - \tau_l). \quad (1)$$

The TDL model does not represent angular or spatial features of the propagation channel. In contrast, the clustered delay line (CDL) model incorporates spatial information and therefore provides a more realistic channel description. Similar to the TDL model, the CDL model defines a set of multipath clusters characterised by relative delays and average powers. In addition, each cluster is associated with angular parameters, including the azimuth and zenith angles of departure and arrival (AoD, AoA, ZoD, and ZoA). Each cluster is further decomposed into multiple sub-rays with random initial phases and Doppler components. By modelling these spatial characteristics, the CDL channel model enables a more accurate representation of the spatial characteristics of MIMO channels, at the cost of increased complexity and higher computational cost [6].

In addition to CSI, 5G NR also employs hybrid automatic repeat request (HARQ) to improve the robustness of the transmission through re-transmissions. After DL-SCH decoding at the receiver, an acknowledge/negative acknowledge (ACK/NACK) signal is fed back to the transmitter, which indicates whether a re-transmission is required. In the case of a decoding failure, the transport block is re-transmitted using a different redundancy version (RV) and combined at the receiver, thus increasing the probability of successful decoding [2] [7].

2.3 CDL Channel Model and Doppler Shift

CDL model definition

CDL ABCDE difference, why CDL-C default

Doppler shift definition and impact

2.4 MATLAB Simulation Framework

Table 2 below summarises the baseline system and channel configuration adopted in this study, which is aligned with the assumptions commonly used in 3GPP evaluation studies, including TR 38.753 [8]. There different mobility scenarios are considered. The low-mobility case assumes a maximum Doppler shift of 10 Hz, corresponding to a UE speed of 3 km/h, a default setting for 3GPP testing. For medium- and high-mobility cases, the 100 Hz and 390 Hz Doppler shift corresponds to the speed of approximately 30 km/h and 120 km/h respectively for a 3.5 GHz carrier frequency.

Table 2: System and channel configuration used in the simulations

Parameter	Value
Carrier frequency	FR1, 3.5 GHz
Subcarrier spacing	30 kHz
Channel bandwidth	106 RBs
Antenna configuration	4 Tx, 4 Rx
Maximum number of layers	4
Channel model	CDL-C
Mobility scenarios	10 Hz (Default), 100 Hz, 390 Hz
Simulation type	Downlink link-level

The CDL-C channel model is selected as it represents a typical non-line-of-sight (NLOS) urban macro environment and is widely used in 3GPP evaluation studies, while modelling spatial characteristics are essential for the study of CSI feedback. A 4×4 MIMO configuration with up to four transmission layers is considered, reflecting a representative frequency range 1 (FR1) deployment while maintaining a manageable simulation complexity. This configuration allows multi-layer transmission as well as rank adaptation to be studied within a single-codeword framework, avoiding additional variability introduced by using an additional codeword, and thus enabling a clearer assessment of the effect of the CSI feedback parameters.

The full set of parameters used are shown in the Appendix section. Fig. 2 shows the downlink resource grid of the system under such configuration. The empty...

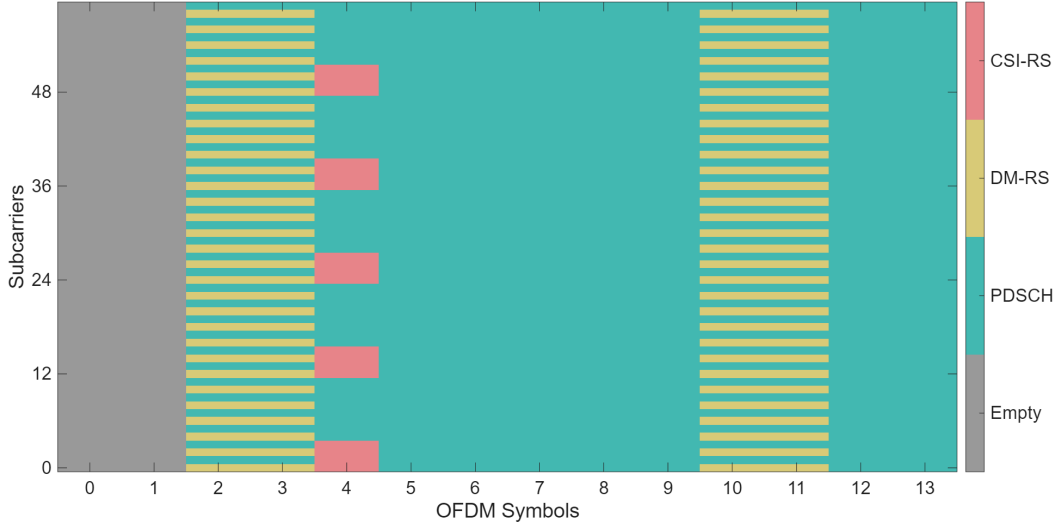


Figure 2: Downlink resource grid showing the allocation of PDSCH, DM-RS and CSI-RS

2.5 System Configuration and Experimental Design

This study focuses on investigating the impact of CSI feedback parameters on downlink transmission performance. The baseline CSI configuration follows the settings specified in relevant 3GPP specifications, including TS 38.101-4 [9]. Both the baseline configuration and the investigated parameter ranges are summarised in Table 3. The investigated ranges represent the set of valid discrete configurations specified by 3GPP. Unless otherwise stated, all simulations use the baseline CSI configuration listed in Table 3.

Table 3: Baseline CSI feedback configuration used in the simulation framework

Parameter	Baseline setting
CSI report periodicity	10 slots
CQI reporting mode	Wideband
PMI reporting mode	Wideband
Subband size	16 RBs
RI configuration	Adaptive
Codebook type	Type I
Codebook mode	Mode 1

In section 3, the behaviour of the system under baseline condition will be simulated and investigated. In section 4, the performance of the system under different MIMO, CDL channel configurations will be investigated, which examines whether the baseline behaviour observed in the baseline scenario remains interpretable under these changes. In section 5, the CSI reporting period will be varied to analyze suitable reporting period for different scenarios and the underlying reasons behind the observations.

Only a subset of the parameters under investigation is varied in each experiment, while the remaining parameters are kept fixed at their baseline values to ensure a controlled comparison. Table 4 shows a brief summary of the simulation study groups that are going to be simulated. Note that sometimes may change multiple parameters at the same time.

Table 4: Summary of simulation study groups and objectives

Study Group	Purpose	Varied parameter
Baseline	CDL-C 4×4 reference performance	—
Fixed-rank controls	Layer adaption	$RI = 1$, $RI \leq 2$, $RI \leq 4$
MIMO configuration sweep	Antenna configuration sensitivity	2×2 , 4×4 , 8×4
CDL profile sweep	Channel-model robustness	CDL-B, CDL-C, CDL-D
Doppler sweep	Mobility sensitivity	10 Hz, 100 Hz, 390 Hz
CSI period sweep	CSI feedback freshness	4, 10, 64, 160 slots

Table 5: Simulation execution settings

Parameter	Value
SNR range	−5 to 35 dB, with 1 dB spacing
Number of simulated frames	1500 10-ms frames per SNR point
Channel estimation	Practical channel estimation

3 Baseline CSI-Feedback Downlink Behaviour

The behaviour of the system under the baseline case is tested in this section. The performance of the system as well as the role of CSI is investigated and analyzed in detail to provide a good understanding of the feedback decisions. At the second half of the section, a fixed-rank control is implemented to isolate the rank adaptation mechanism and .. the multi-layer transmission.

3.1 Baseline CDL-C 4×4 Throughput Behaviour

Fig. 3 presents the baseline downlink throughput for the CDL-C 4×4 MIMO configuration with adaptive CSI feedback. At the simulated range of -5 dB to 25dB, the throughput curve is increasing. The curve shows a gradual increase at low SNR, followed by a steeper transition region between approximately 5 dB and 17 dB. At 10dB, the throughput is approximately xxx Mbps. The throughput saturates at high SNR of approximately xxx Mbps. The behaviour of the system is explained in the next subsection according to the CSI-related behaviours. This baseline provides the reference case against other scenarios compared in the subsequent simulations.

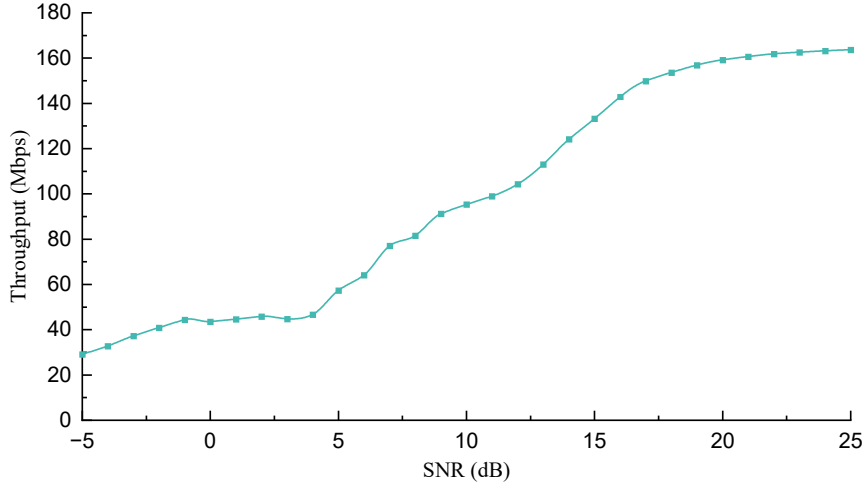


Figure 3: Baseline downlink throughput performance for the CDL-C 4×4 configuration with adaptive CSI feedback

3.2 CSI-driven Link Adaptation Statistics

In addition to the performance of the system, the behaviour of the CSI during the transmission is shown in Fig. 4 below. The four respective figures demonstrates the changes of CQI, target code rate (TCR), RI as the SNR changes as well as the use of modulation schemes at different SNR levels. CQI, TCR and RI are not fixed for each SNR testing point, but instead varies base on the real-time feedbacks and channel condition, even within subband (?). vertical bars which functions as the distribution spread indicator has been introduced to the three subfigures to characterize the consistency and variability. The bars indicate the 5th to 95th percentile range of each value range rather than the full minimum–maximum range. This choice is used to represent the typical spread of the selected CQI values while reducing the influence of rare extreme channel deep-fades or favorable fading realizations, making the results more representative.

For CQI plot, as the SNR increases, the selected CQI rises gradually from 5 at low SNR to around 11 at high SNR, indicating that the link adaptation gets gradually more aggressive during the transmission due to the improved channel conditions. The peak CQI the system has every selected is 14.

Correspondingly, the target code rate of the system follows a similar trend, overall increasing from ... to nearly The TCR curve is not strictly increasing but instead having a oscillatory behaviour. The local minimum occur at SNR points where major change in modulation scheme occurs as the system would select a higher modulation scheme but in the meantime using a lower TCR. The two major local minimum occurs at around -3 dB and 4 dB, which corresponds to the SNR when the system switches the main modulation from QPSK to 16QAM, and from 16QAM to 64QAM respectively.

The selected RI also changes with SNR, reflecting that the system changes the number of layers used for transmission under different channel conditions. At SNR lower than 2 dB, the system mainly selected $RI = 1$, representing a robust single-layer transmission. The RI increases to 2 at higher SNR, enabling steeper increase in throughput. Since the MIMO configuration is 4×4 , a maximum of 4-layer transmission can be supported. However, the RI does not increase beyond 2 under this scenario, likely due to, partly causing the throughput to saturate to a throughput level below theoretical limit at very high SNR.

For table 1 and 3 (...), three modulation schemes are supported, which are QPSK, 16QAM and 64QAM. 256 QAM is only supported when (...) and is uncommonly used due to its high requirement on channel condition. At very low SNR, the main modulation scheme is QPSK. As the SNR increases, QPSK quickly disappears and 16 QAM becomes dominant. For SNR higher than 10dB, 64QAM modulation dominates. The change in modulation scheme is gradual instead of abrupt, where the higher order modulation scheme will gradually be used more frequently as the SNR increases. The system uses 64 QAM for around 90% of the cases, while the rest of the time using 16 QAM, likely during unusual channel degradation such as deep fade.

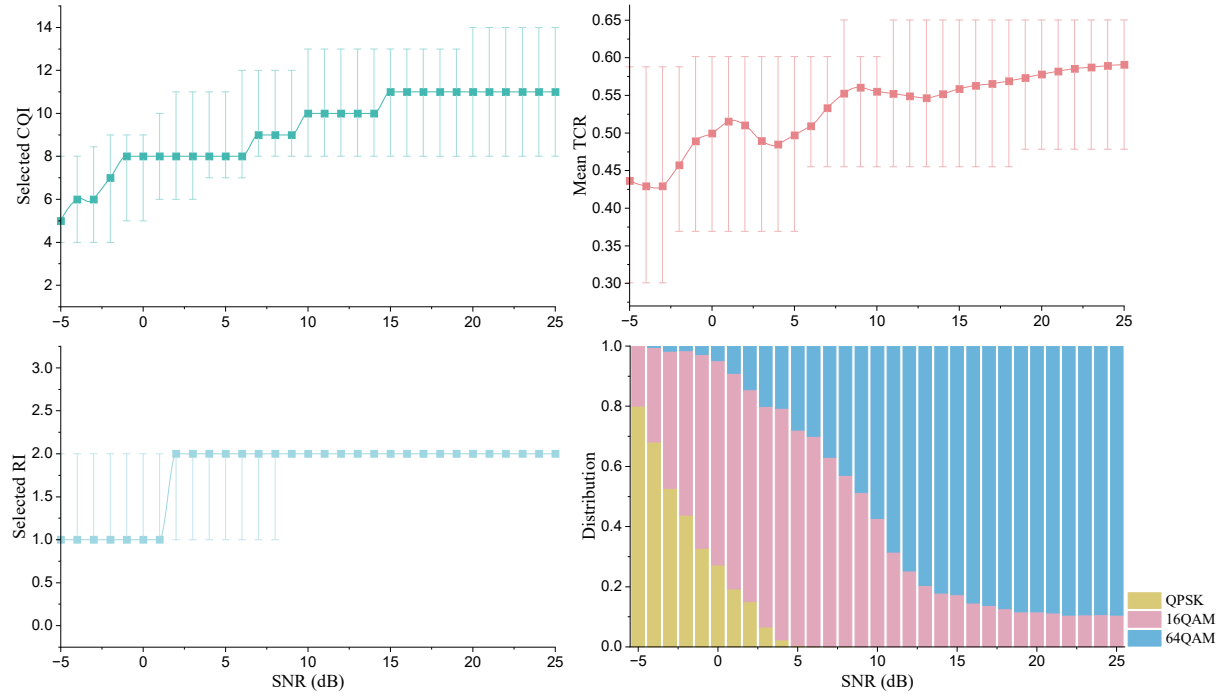


Figure 4: CSI-driven link adaptation behaviour across SNR

Explain why shape of throughput curve

3.3 Fixed-Rank Controls

Figure 5 compares the downlink throughput under different rank-restriction configurations. The $RI \leq 4$ case corresponds to the original baseline, where up to four transmission layers are permitted. The $RI \leq 2$ case restricts the adaptive RI selection to at most two layers, while the $RI = 1$ case forces single-layer transmission.

The $RI \leq 4$ and $RI \leq 2$ curves are almost identical across the full SNR range. This confirms that, under the configuration considered here, allowing more than two layers provides little additional throughput benefit. This observation is consistent with the link adaptation statistics in Figure 4, where the selected RI increases from one to two at low-to-moderate SNR but does not increase beyond $RI = 2$.

The fixed $RI = 1$ curve behaves differently. At low SNR, single-layer transmission can be slightly more robust than two-layer transmission, leading to lower BER and therefore higher throughput. However, as SNR increases, the lack of spatial multiplexing becomes the dominant limitation. The $RI = 1$ throughput saturates at around xxx Mbps starting from around 17 dB, whereas the adaptive RI cases reach approximately xxx Mbps. This shows that adaptive multi-layer transmission is essential for achieving the high-SNR throughput gain. The higher-rank transmission beyond two layers is not effectively exploited in this baseline scenario, but the usage of adaptive layer transmission will be even bigger in a richer $\mathcal{H}$ channel that supports more independent paths $\mathcal{H}$.

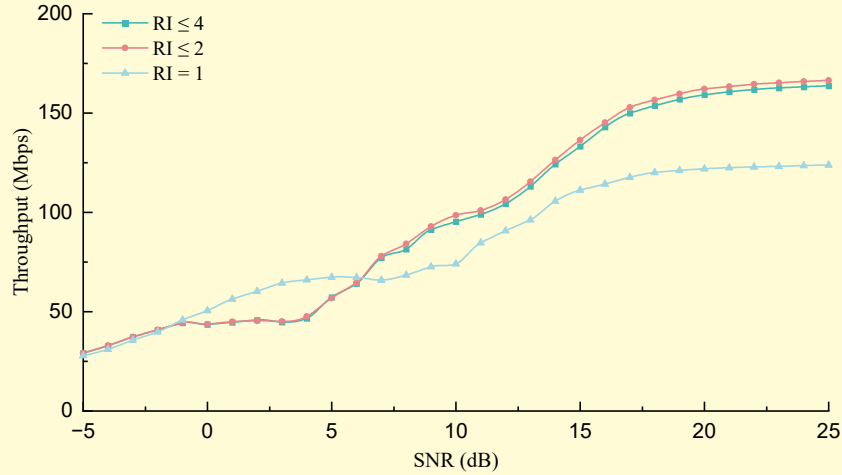


Figure 5: Downlink throughput performance under different maximum layer configurations

3.4 Summary of Baseline Findings

1. The above result is the baseline
2. Examines the impact of CSI and how it actually works

4 Robustness across MIMO and Channel Configurations

This chapter examines if the baseline behaviour observed in the previous section remains consistent under different MIMO and CDL channel configurations. The aim of this part is to check whether the main link-adaptation trends remain interpretable when the spatial and channel conditions are changed.

4.1 Antenna Configuration Sweep

Figure 6 compares the downlink throughput performance under different MIMO antenna configurations. The 4×4 case corresponds to the baseline configuration discussed in the previous chapter. Two additional configurations 2×2 and 8×4 are tested.

Despite the fact that, as also shown in the previous section that the maximum number of layers that are actually been used are 2, having a larger antenna array leads to better performance of the system due to

The 2×2 configuration leads to a noticeably lower throughput across the SNR ranges as compared to the baseline 4×4 configuration and saturates at around 90 Mbps. This is as expected, as a smaller antenna configuration leads to fewer spatial degrees of freedom.

Oppositely, the 8×4 case can in general lead to better performance of the system. At low-to-medium SNR region, the increase in Tx antennas leads to higher throughput than the baseline, suggesting that the larger transmit array improves the effective channel quality and allows the link adaptation algorithm to select more efficient transmission parameters at lower SNR values ρ_{eff} . The throughput for baseline outperforms 8×4 , however, when the SNR is above 17dB, reaching approximately xxx Mbps compared with around xxx Mbps for 8×4 .

This difference should not be interpreted as evidence that 8×4 MIMO is inherently worse at high SNR. The saturated performance of the system is a result of the CSI feedback, which often lead to slightly different CQI, thus different TCR, layer or even transmission schemes. The result here indicates that the additional transmit antennas are not fully converted into a higher peak throughput under the current CSI feedback configuration. In this setup, the benefit of 8×4 is mainly observed as an SNR gain of approximately 5 dB in the low-to-medium SNR region rather than as a higher saturation throughput.

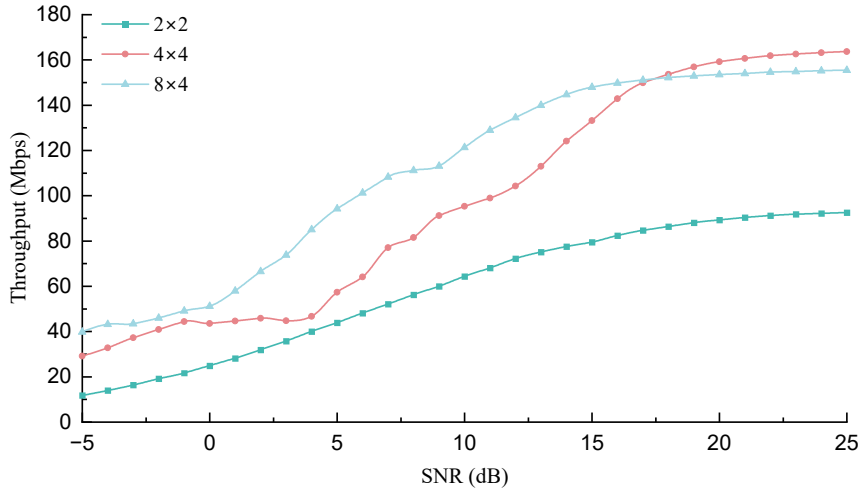


Figure 6: Downlink throughput performance under different MIMO configurations **Still need to check 1x1 and 4x8 and 8x8? results**

4.2 CDL Channel Profile Sweep

The second check varies the CDL channel profile while keeping the baseline 4×4 MIMO configuration and CSI feedback settings unchanged.

Figure 7 compares the downlink throughput performance under CDL-B, a typical NLOS channel, and CDL-D, a typical LOS channel with the baseline. The CDL-C case corresponds to the baseline configuration. ?????Maybe explain why CDL-C is most commonly used.?????

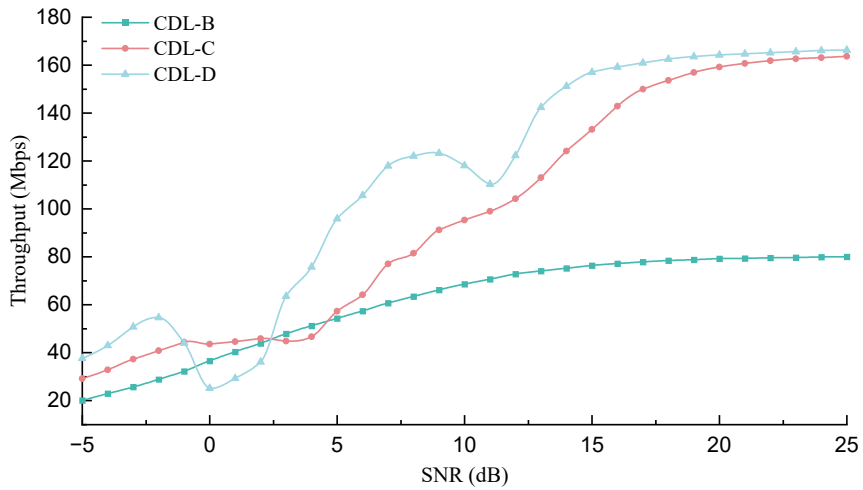


Figure 7: Downlink throughput performance under different CDL channel models

For all three CDL profiles, the throughput increases with SNR and eventually approaches saturated levels, showing that the general link-adaptation behaviour remains consistent.

However, the performance of the system under different channel models vary greatly. The CDL-B profile increases gradually, almost always worse than the baseline scenario and saturates at a much lower throughput, around xxx Mbps, as compared to baseline CDL-C. This is as expected as the CDL-B channel model has lower potential in spatial multiplexing as compared to CDL-C.

The CDL-D profile on the other hand achieves the best overall performance. It generally leads to higher throughput at low-to-medium SNR regions and, while leading to almost the same saturated throughput as CDL-C at the high SNR, enters the saturated zone in advance. The overall performance has approximately 5 dB in SNR gain as compared to the baseline. This behaviour is also as expected, as the CDL-D channel has a major LOS component, which supports reliable and rapid transmission even at a low SNR situation, leading to clear advantage as compared to other scenarios, when one-layer transmission is used.

Overall, the CDL profile sweep shows that regardless the channel used, the overall shape of the throughput curve is to a large extent consistent. But the achievable throughput and required SNR are channel-dependent. For example, among the two additional models tested here, CDL-D mainly improves the transition-region SNR requirement while CDL-B represents a more limiting channel condition under the current setup.

4.3 One-layer Control Experiments

Similar to section 3.3, a controlled experience is conducted to test the performance of the simulations in this section performs when the maximum number of layers supported is fixed to 1, as shown in Figure 8. In contrast to the adaptive baseline, where the rank indicator is selected through CSI feedback, the transmission rank is constrained to one layer throughout the SNR sweep.

Considering the 4×4 cases, the conclusion from Fig. 7 remains mostly unchanged. CDL-B model is, across the whole simulated SNR range, largely worse than the baseline scenario. CDL-D channel outperforms the baseline at low-and-medium SNR regions, while achieving slightly lower saturated throughput than the baseline.

Compared with the adaptive CDL profile sweep, the fixed one-layer result provides a useful diagnostic reference. In the adaptive case, performance differences may arise from both single-layer link quality and the ability of the channel to support spatial multiplexing. Under fixed one-layer transmission, however, rank adaptation is removed. Therefore, the remaining differences mainly reflect how each CDL profile supports robust single-layer transmission under the same CSI feedback and other configurations. Considering the sweep of MIMO configurations when the CDL-C channel model is used, the

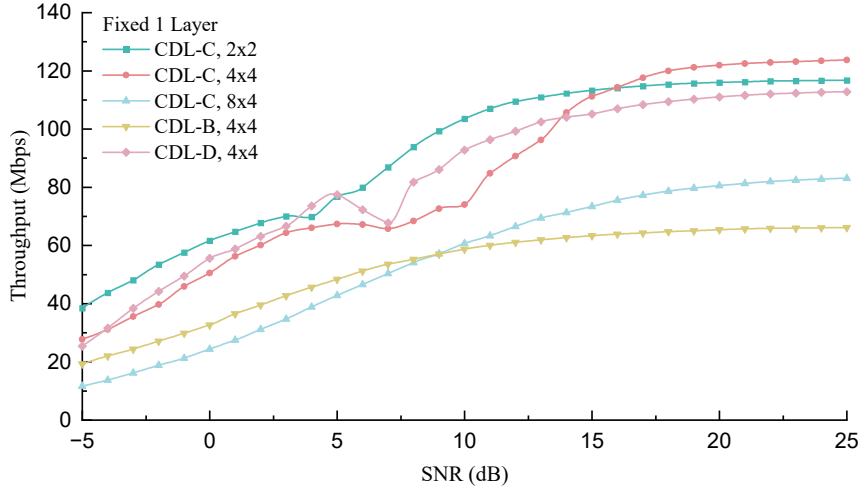


Figure 8: Downlink throughput comparison under fixed one-layer transmission Some issue with [2x2] and [8x4] curves, re-simulate

5 Impact of Mobility and CSI Reporting Periodicity

This chapter investigates the impact of mobility and CSI reporting periodicity on the CSI-driven downlink transmission performance. In a time-varying channel, the CSI obtained from CSI-RS measurements may become outdated before it is applied for PDSCH transmission. This CSI ageing effect is expected to become more significant as the Doppler shift increases or as the CSI reporting period becomes longer. Therefore, this chapter first studies the impact of Doppler shift under the baseline CSI reporting period, before examining the effect of different reporting periodicities.

5.1 Doppler Shift under Default CSI Reporting Period

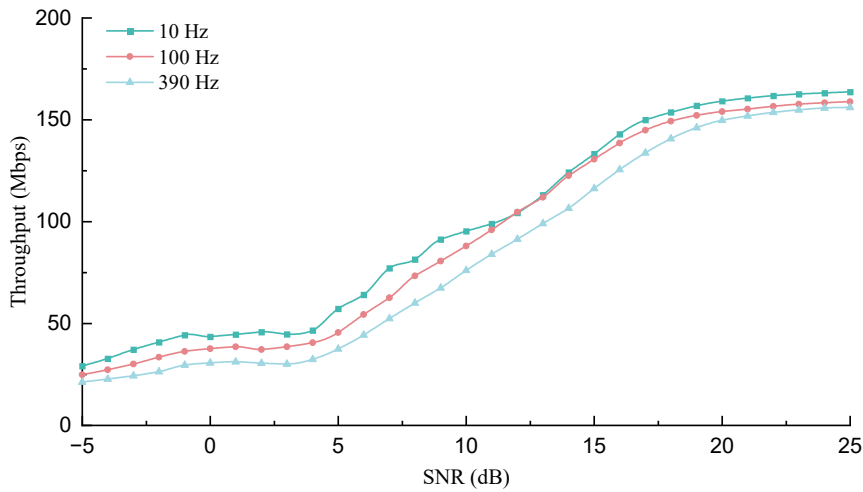


Figure 9: Impact of Doppler shift on downlink throughput

Table 6: Interpolated SNR values at selected target throughput levels

Target Throughput (Mbps)	SNR 10 Hz (dB)	SNR 100 Hz (dB)	SNR 390 Hz (dB)
50	4.31	5.50	6.69
80	7.66	8.91	10.50
110	12.66	12.73	14.35
140	15.70	16.23	17.89

[†] SNR values obtained using linear interpolation.

Figure 9 shows the impact of Doppler shift on downlink throughput under the baseline CSI reporting period. The three Doppler shifts, 10 Hz, 100 Hz, and 390 Hz, represent increasing levels of channel time variation. Using a reference carrier frequency of 3.5 GHz, these correspond approximately to UE speeds of 3 km/h, 31 km/h, and 120 km/h, respectively.

The general throughput behaviour remains similar across all Doppler cases: the throughput increases with SNR, enters a transition region, and eventually approaches a high-SNR saturation level. However, higher Doppler shift causes a visible rightward shift of the throughput curve. This indicates that, under higher mobility, a larger SNR is required to achieve the same throughput level.

This observation is quantified in Table 6, where the SNR values required to reach selected target throughput levels are obtained by linear interpolation. For example, to reach 80 Mbps, the required SNR increases from 7.66 dB at 10 Hz to 8.91 dB at 100 Hz and 10.50 dB at 390 Hz. This corresponds to an SNR penalty of approximately 1.25 dB and 2.84 dB relative to the 10 Hz baseline. A similar trend is observed at 140 Mbps, where the 390 Hz case requires 17.89 dB compared with 15.70 dB for the 10 Hz case.

The degradation is consistent with CSI ageing. With a fixed CSI reporting period and processing delay, the CSI used for link adaptation becomes less representative of the actual channel at the time of transmission when the channel varies more rapidly. As a result, the selected RI, PMI, and CQI may become less accurate, leading to less effective precoding and MCS selection. The effect is most visible in the transition region, where link adaptation decisions are particularly sensitive to the estimated channel quality.

At high SNR, the curves become closer and approach similar saturation levels, although the higher-Doppler cases remain slightly lower. This suggests that high SNR can partially compensate for mobility-induced CSI mismatch, but does not fully remove the loss caused by outdated CSI. These results motivate the following periodicity study, where shorter CSI reporting periods are examined as a potential way to improve CSI freshness under higher mobility.

These results motivate the following CSI periodicity study. Since the degradation at higher Doppler is partly caused by CSI ageing, reducing the CSI reporting period may help improve the freshness of the feedback and recover part of the throughput loss under high-mobility conditions.

5.2 Reporting Periodicity at Low and High mobility

Figure 10 compares the effect of CSI reporting periodicity under low- and high-mobility scenarios. The 10 Hz case represents a slowly varying channel, while the 390 Hz case represents a high-mobility channel with stronger time variation. In both cases, the CSI-RS and CSI reporting periods are varied while the remaining baseline configuration is kept unchanged.

For the 10 Hz case, the throughput curves are relatively close across different reporting periods. Shorter reporting periods, such as 4 and 10 slots, provide slightly better performance in the transition region, while longer reporting periods lead to a small rightward shift of the curve. This indicates that, under low mobility, the channel changes slowly enough that CSI remains useful even when the reporting interval is increased. As a result, reducing the reporting period provides only limited throughput improvement.

For the 390 Hz case, the impact of reporting periodicity is more visible. The 4-slot reporting period achieves the best performance in the transition region, while longer periods require a higher SNR to reach similar throughput levels. This is consistent with the CSI ageing effect: when the channel varies rapidly, CSI obtained from earlier CSI-RS measurements becomes outdated more quickly, reducing the accuracy of RI, PMI, and CQI selection. More frequent reporting helps improve CSI freshness and therefore improves link adaptation under high mobility.

However, the difference between reporting periods remains relatively modest in the high-SNR region. All curves eventually approach a similar saturation throughput, suggesting that CSI reporting periodicity mainly affects the SNR required to enter the high-throughput region, rather than the final saturation throughput itself. Therefore, the benefit of shorter CSI reporting is most relevant in the transition region and under high-mobility conditions.

Overall, the results show that reducing the CSI reporting period provides limited benefit under low mobility, but becomes more useful under high mobility where CSI ageing is more severe. The main effect is an SNR gain in the transition region rather than a significant increase in high-SNR saturation throughput.

This also highlights a practical trade-off. More frequent CSI reporting can improve the freshness of CSI and slightly improve throughput under high mobility, but it also increases uplink feedback overhead. Therefore, an excessively short reporting period may not be necessary in low-mobility scenarios, where the throughput gain is small.

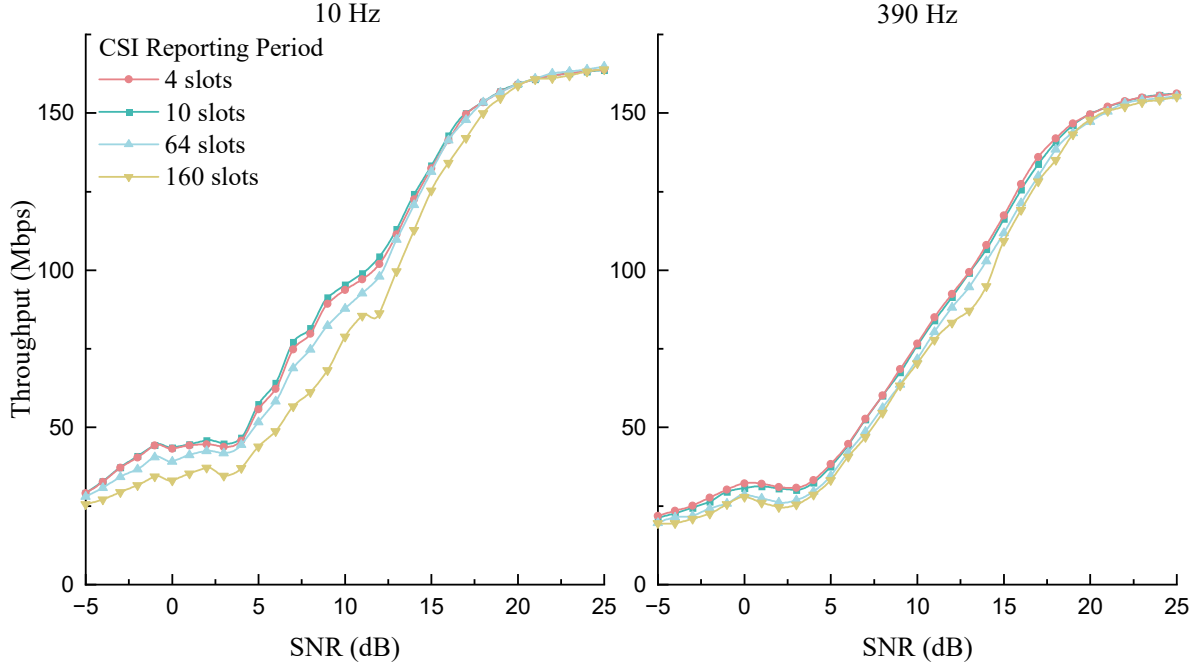


Figure 10: Downlink throughput performance under different CSI reporting periods for low- and high-mobility scenarios

A CSI reporting period of 10 slots is used as the baseline, denoted as $P_0 = 10$. For each Doppler shift f_D , the reference peak throughput is defined as

$$R_{f_D, P_0}^{\max} = \max_{\gamma} R_{f_D, P_0}(\gamma), \quad (2)$$

where $R_{f_D, P}(\gamma)$ is the measured downlink throughput at SNR γ under CSI reporting period P . The target throughput level used for comparison is 50% of the reference peak throughput is $0.5R_{f_D, P_0}^{\max}$.

Therefore, the SNR required for each reporting period to reach the target throughput is obtained by linear interpolation as $\gamma_{f_D, P}(0.5R_{f_D, 10}^{\max})$. The relative SNR penalty or gain with respect to the baseline CSI report is then

$$\Delta\gamma_{f_D, P} = \gamma_{f_D, P}(0.5R_{f_D, 10}^{\max}) - \gamma_{f_D, 10}(0.5R_{f_D, 10}^{\max}). \quad (3)$$

Positive values indicate an SNR penalty, while negative values indicate an SNR gain.

Table 7: SNR required to reach 50% of the baseline reference peak throughput

Doppler	CSI period	$\gamma_{50\%}$	$\Delta\gamma$
10 Hz	4 slots	8.2 dB	0.1 dB
10 Hz	10 slots	8.1 dB	–
10 Hz	64 slots	9.0 dB	0.9 dB
10 Hz	160 slots	10.5 dB	2.4 dB
100 Hz	4 slots	8.5 dB	-0.5 dB
100 Hz	10 slots	9.0 dB	–
100 Hz	64 slots	11.0 dB	2.0 dB
100 Hz	160 slots	10.8 dB	1.8 dB
390 Hz	4 slots	10.3 dB	-0.1 dB
390 Hz	10 slots	10.4 dB	–
390 Hz	64 slots	10.8 dB	0.4 dB
390 Hz	160 slots	11.2 dB	0.8 dB

[†] Reference peak throughput values used: 163.9 Mbps at 10 Hz, 161.0 Mbps at 100 Hz, 158.0 Mbps at 390 Hz.

Table 7 quantifies the SNR required to reach 50% of the baseline reference peak throughput under each Doppler condition. Since the reference peak throughput is defined separately for each Doppler shift, the values should be interpreted as the relative impact of CSI reporting periodicity within the same mobility condition, rather than as a direct comparison of absolute throughput across different Doppler shifts.

At 10 Hz, changing the reporting period has only a limited effect when the period is short. The 4-slot and 10-slot cases require almost the same SNR to reach the target throughput, with only a 0.1 dB difference. However, increasing the reporting period to 64 and 160 slots introduces clear SNR penalties of 0.9 dB and 2.4 dB, respectively. This indicates that although the channel is slowly varying at 10 Hz, very infrequent CSI reporting can still degrade link adaptation because the feedback becomes less representative of the current channel state.

At 100 Hz, the impact of reporting periodicity becomes more visible. The 4-slot case achieves a 0.5 dB SNR gain relative to the 10-slot baseline, showing that more frequent reporting can improve CSI freshness when the channel varies more rapidly. In contrast, the 64-slot and 160-slot cases require approximately 2.0 dB and 1.8 dB additional SNR, respectively. The non-monotonic difference between 64 and 160 slots should not be over-interpreted, as it may result from finite simulation length, interpolation error, and the discrete nature of CSI-driven CQI, PMI, and RI decisions. The main observation is that long reporting periods lead to a clear SNR penalty compared with the 10-slot baseline.

For the 390 Hz case, the SNR penalties are smaller at this 50% reference-throughput point. The 4-slot and 10-slot cases are almost identical, while the 64-slot and 160-slot cases introduce penalties of 0.4 dB and 0.8 dB. This suggests that, under very high mobility, shortening the reporting period alone does not fully recover the loss caused by rapid channel variation. The CSI is affected not only by the reporting period, but also by processing delay, channel estimation accuracy, and the time variation between CSI acquisition and PDSCH transmission.

Overall, the results show that CSI reporting periodicity mainly affects the SNR required in the transition region rather than the final saturation throughput. Shorter reporting periods can provide an SNR gain by reducing CSI ageing, especially under moderate mobility. However, the gain is not large in all cases and must be balanced against the additional uplink feedback overhead. Therefore, a very short reporting period may not be necessary under low mobility, while very long reporting periods can noticeably degrade link adaptation performance.

The main impact of CSI reporting periodicity is therefore an SNR penalty or gain in the transition region, rather than a significant change in the high-SNR saturation throughput.

5.3 Discussion of CSI Freshness and Practical Trade-Off

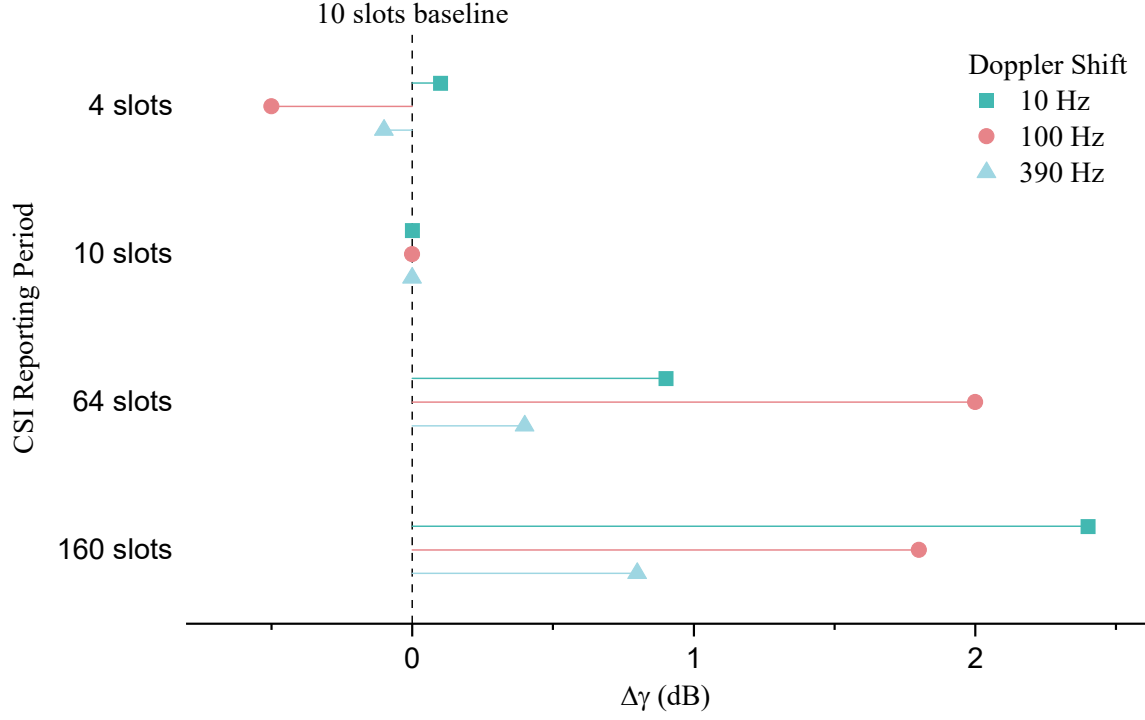


Figure 11: Relative SNR penalty/gain $\Delta\gamma$ for different CSI reporting periods and Doppler shifts (The 10-slot reporting period is used as the baseline for each Doppler shift)

Summarize 4.2 and 4.3 CSI feedback should not be evaluated only by its granularity or reporting frequency. Its value depends on whether the reported channel state remains relevant over the time scale of channel variation. 1. What exactly is the cost of CSI feedback 2. Under what scenarios are frequent CSI report actually useful

The results in Sections ?? and ?? show that the value of CSI feedback should not be evaluated only by its reporting granularity or reporting frequency. Instead, its effectiveness depends on whether the reported channel state remains sufficiently fresh over the time scale of channel variation. When the channel varies slowly, as in the 10 Hz Doppler case, the CSI remains relevant for a longer period. Therefore, reducing the reporting period from 10 slots to 4 slots provides only a small throughput improvement. In contrast, very long reporting periods can still introduce a visible SNR penalty because the feedback becomes less representative of the current channel state.

Under higher mobility, CSI ageing becomes more important. The Doppler sweep shows that higher Doppler shifts lead to a rightward shift of the throughput curve, meaning that a larger SNR is required to achieve the same throughput. This degradation is consistent with the reduced accuracy of CSI-driven RI, PMI, and CQI selection when the channel changes between CSI acquisition and data transmission. Shorter reporting periods can partly mitigate this effect by improving CSI freshness, especially in the transition region where link adaptation decisions are most sensitive to channel quality. However, the periodicity results also show that shorter CSI reporting is not universally beneficial. At high SNR, the throughput curves for different reporting periods converge to similar saturation levels. This suggests that reporting periodicity mainly affects the SNR required to enter the high-throughput region, rather than the final saturation throughput. Moreover, under very high Doppler shift, reducing the reporting period alone cannot fully compensate for rapid channel variation, since processing delay, channel estimation accuracy, and the time variation between CSI-RS measurement and PDSCH transmission also contribute to CSI mismatch.

This leads to a practical trade-off. More frequent CSI reporting improves the freshness of CSI and can reduce the SNR required to achieve a given throughput, but it also increases uplink feedback overhead. This overhead includes the resources required to transmit RI, PMI, and CQI reports, and becomes more significant when reporting is frequent or when subband feedback is used. Therefore, an excessively short reporting period may be inefficient in low-mobility scenarios, where the throughput gain is small. Conversely, very long reporting periods can degrade link adaptation when mobility is moderate or high.

Overall, the results suggest that CSI reporting should be configured according to the mobility condition and the expected channel coherence time. A moderate reporting period may be sufficient for low-mobility users, while shorter periods become more valuable when CSI ageing is significant. The main benefit of more frequent reporting is therefore an SNR gain in the transition region, rather than a substantial increase in the high-SNR peak throughput.

6 Discussion and Conclusion

6.1 Interpretation of Results

6.2 Limitations

6.3 Future Work

7 References

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8 Appendix

8.1 Risk Assessment

A complete risk assessment was provided in the submitted Technical Milestone Report (TMR). Since this project is entirely simulation-based and does not involve any physical laboratory experiments, no safety-related or hardware-related risks were identified. The only practical risk identified was the limited time available for the project relative to the computational cost of the simulations.

The risk assessment made in the TMR remains valid, and the identified computational risk has been carefully managed throughout the project. For example, during the development of the simulation program, specific care was taken to ensure that the code could be executed in parallel on multiple CPU workers using the MATLAB Parallel Computing Toolbox, rather than being limited to a single CPU thread at a time. In addition, external computing resources were used where necessary, including the CUED central computing system, `ts-access`.

8.2 Ethical Statement

This project is entirely simulation-based and does not involve personal data or physical laboratory experiments. Therefore, no direct safety, privacy, or human-related ethical risks are identified.

From an environmental perspective, the project requires computational resources for MATLAB simulations. Although the total computational demand is very limited relative to most other simulation-based research projects, unnecessary computation was avoided where possible by using a consistent baseline setup, a clearly pre-designed experimental structure and selecting an appropriate number of simulated frames to balance result accuracy and computational cost.

Additionally, the conclusion of the project may have positive social and environmental implications, as it provides a detailed analysis of CSI feedback and MIMO techniques, which could potentially contribute to more efficient wireless communication system. For example, it may help avoid using overly frequent uplink reports when unnecessary. Improved link adaptation can in the meantime support better spectral efficiency and more reliable connectivity.

8.3 Declaration of the use of generative AI

Generative AI tools, specifically ChatGPT and Google Gemini, were used in a limited number of situations during the completion of the project. The AI assistance is used mainly during the early stage of the project, including to facilitate the understanding of related concepts, as well as the debugging of the MATLAB simulation implementation. It can be confirmed that the experimental design, data analysis, interpretation of results, final conclusions, and report content were completed and verified by the author only. No AI tool was used, in any form, to generate, modify, or manipulate the simulated data.

8.4 Detailed Baseline Simulation Configuration

Table 8: Baseline simulation parameters, adapted from TR 38.753 where applicable [8]

Parameter		Value
Duplex mode		TDD
FR / Carrier frequency		FR1, 3.5GHz
Maximum Doppler shift		10 Hz
Channel Bandwidth / SCS		40Mhz / 30kHz
Maximum allowed rank		4
Antenna configuration		4Tx4Rx
Channel model		CDL-C
Delay spread		300 ns
Codebook configuration for PDSCH and DMRS		Single Panel Type I
MCS table		Table 1 (64 QAM)
PDSCH configuration	Mapping type	Type A
	k0	0
	Starting symbol	2
	Length	12
PDSCH DMRS configuration	DMRS Type	Type 1
	Number of additional DMRS	1
	Max. number of OFDM symbols for DL front loaded DMRS	2

Table 9: Baseline CSI simulation parameters, adapted from TR 38.753 where applicable [8]

Parameter		Unit	Value
NZP CSI-RS for CSI acquisition	CSI-RS resource type		Periodic
	Density		1
	Symbol location		4
	Subcarrier location		0
	CSI-RS interval and offset	slot	10, 1
CQI			Wideband
PMI			Wideband
CQI/RI/PMI delay		ms	7

8.5 Project Repository and Supporting Materials

The simulation code and supporting materials developed for this project are made available in a GitHub repository: <https://github.com/Bariumoxid/MIMOTechniques>. The repository contains the main MATLAB simulation scripts, configuration files, workflows, and main test cases used to generate and validate the results presented in this report. Other files, such as the weekly progress reports, log book as well as the raw files for the reports and graphics were also available on the site.